

Restaurant Delivery Profitability & Competitor Intelligence System Egyptian Market Edition

1. Project Overview:

This project delivers a complete Business Intelligence system for Egyptian restaurants to optimize delivery profitability. It tracks orders across Talabat, Elmenus, and WhatsApp, calculates real-time profit per order, predicts COD refusal risk, and monitors competitor prices weekly. The system includes a SQL Server database, stored procedures, SSRS reports, a star schema data warehouse, and 20 Power BI dashboards with "Kill or Keep" menu recommendations. AI agents via n8n automate WhatsApp alerts for high-risk orders and daily profit reports, reducing losses by up to 50%.

2. Business Understanding

The Egyptian food delivery market is growing rapidly, but restaurants face major challenges including high platform commissions, COD refusals, lack of competitor visibility, fluctuating ingredient costs, and poor menu optimization decisions. These issues limit profitability and operational efficiency, highlighting the need for a data-driven system that enables better visibility, smarter decisions, and sustainable growth.

3. Project Objectives

This project aims to empower Egyptian restaurants to optimize their delivery operations by transforming fragmented and opaque data into actionable, profit-driven insights.

The system is designed to provide full visibility across all delivery channels, enabling restaurant owners to accurately track profitability at every level—order, item, platform, and time—while identifying hidden losses caused by high platform commissions.

It also enhances decision-making by predicting high-risk Cash-on-Delivery (COD) orders before dispatch, reducing refusal rates and minimizing operational waste.

In addition, the solution introduces competitive intelligence by continuously monitoring rival restaurants' pricing, promotions, and delivery performance, allowing businesses to remain strategically positioned in the market.

To address cost volatility, the system tracks daily ingredient price fluctuations and evaluates their direct impact on menu profitability, enabling proactive pricing and promotion adjustments.

Finally, the platform supports data-driven menu optimization by analyzing performance metrics and generating clear recommendations on which items to retain, improve, or remove—ensuring a more profitable and efficient delivery menu.

4. Reports

1. Zone Performance Report.
2. Customer Profitability Report.
3. Platform Performance Report.
4. Competitor Comparison Report.
5. Order Details Report.
6. Order + Customer Payment Details Report.

5. Dashboards

1. Executive Command Center.
2. COD Risk & Refusal Analyzer.
3. Platform Battle.
4. Hourly Profit Heatmap.
5. Menu Item Profitability Matrix.
6. Competitor Price Map.
7. Competitor Menu Strategy.
8. Competitor Promo Tracker.
9. Delivery Speed Benchmark.
10. Customer Segmentation.
11. Ingredient Cost Volatility.
12. WhatsApp Order Analytics.
13. Ramadan & Holiday Analyzer.
14. Driver Performance Scorecard.
15. Marketing ROI Dashboard.
16. Customer Churn Analysis.
17. Refund & Order Accuracy.
18. "Kill or Keep" Menu Optimizer.
19. What-If Scenario Simulator.
20. Weekly Action Report (Export).

6. AI Features

COD Risk Monitor

1. Check SQL Server for high-risk COD orders every 15 minutes.
2. Use AI to decide if an alert is needed.
3. Send WhatsApp alert to the restaurant owner.